The Picton and Varty Lake ultramafic dikes: Jurassic magmatism in the St. Lawrence Platform near Belleville, Ontario

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The Picton and Varty Lake ultramafic dikes that intrude Ordovician limestone near Belleville, Ontario, consist of olivine or serpentinized pseudomorphs and Ti-phlogopite phenocrysts set in a groundmass of spinel, perovskite, serpentine, calcite, apatite, and chlorite. Phlogopite exhibits a complex chemical variation and late-stage Ba enrichment (up to 6.8 wt.% BaO). Spinels are chemically zoned and commonly exhibit an atoll texture in which a magnesian–aluminous chromite core is separated by a silicate zone from late-stage magnesian–aluminous titanomagnetite chemically similar to kimberlitic spinel. Atoll spinels from the Picton dike also have a zone of titanian pleonaste surrounding a central grain of magnesian–aluminous chromite at the centre of the atoll structures. Both dikes are extremely silica undersaturated, with high K/Na and Mg/(Mg + Fe^{2+}) ratios. Petrographic and chemical features of the Picton and Varty Lake dikes preclude simple classification, as both dikes have aspects indicative of kimberlitic and lamprophyric affinities. K–Ar age determinations, indicating a Jurassic age of emplacement for both dikes, are supported by paleomagnetic data from the Picton dike. The Varty Lake dike yielded no stable paleomagnetic remanence directions; limestone adjacent to the dike retains a Paleozoic remanence possibly indicating a very low emplacement temperature. The emplacement of both dikes may be related to Grenville-age basement faults reactivated in Jurassic time.

Les dykes ultramafiques de Picton et du lac Varty faisant intrusion dans le calcaire ordovicien près de Belleville, Ontario, sont constitués d'olivine ou de pseudomorphes serpentinisés et de phénocristaux de phlogopite titanifère fixés dans une matrice de spinelle, pérovskite, serpentine, calcite, apatite et chlorite. La phlogopite possède une composition chimique variable complexe et montre en dernier stade un enrichissement en Ba (jusqu'à 6,8% par poids de BaO). Les spinelles sont zonés chimiquement et présentent fréquemment une structure en atoll où un noyau de chromite alumino-magnésienne est isolé par une zone de silicates de la titano-magnétite alumino-magnésienne du stade final de cristallisation et dont la composition chimique est analogue au spinelle kimberlitique. Les spinelles des atolls du dyke de Picton possèdent également une zone de pléonaste titanifère entourant un grain central de chromite alumino-magnésienne au milieu des structures en atoll. Les deux types de dyke sont fortement sous-saturés en silice et leurs rapports K/Na et Mg/(Mg + Fe^{2+}) sont élevés. Les particularités pétrographiques et chimiques des dykes de Picton et du lac Varty ne permettent pas une classification simple car les deux dykes exhibent des affinités kimberlitiques et lamprophyriques. Des datations au K/Ar indiquent un âge jurassique des mises en place des deux dykes corroboré par des mesures paléomagnétiques sur le dyke de Picton. Le dyke du lac Varty n'a pas fourni de directions stables de rémanence paléomagnétique; le calcaire adjacent retient une rémanence paléomagnétique indiquant possiblement une température de mise en place très basse. La mise en place des deux dykes peut être reliée à des failles du socle du Grenville réactivées au Jurassique.

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Introduction

An increasing number of post-Paleozoic ultramafic rocks have been described west of the Appalachian fold belt. These occurrences include: Ile Bizard, Quebec (Clark 1972; Raeside and Helmstaedt 1982), Ithaca, New York (Martens 1924; Watson 1967; Zartman *et al.* 1967; Kay *et al.* 1983), Dixonville and Masontown, Pennsylvania (Deines 1969; Pimentel *et al.* 1975; Meyer 1976; Hunter and Taylor 1984; Hunter *et al.* 1984), Elliott County, Kentucky (Zartman *et al.* 1967; Garrison and Taylor 1980), and Norris Lake, Tennessee (Meyer 1975).

The spatial distribution and emplacement ages of these kimberlites and related igneous rocks have been used to formulate tectonic models relating to their emplacement. These models include relating their emplacement to the fossil trace of a hot spot (Crough 1981) or to the reactivation of faults during the reopening of the Atlantic during the Jurassic (Parrish and Lavin 1982). Ultramafic dikes of post-Ordovician age have been recognized at two localities north of Lake Ontario (Fig. 1). One of the dikes, discovered by C. G. Winder, is exposed in a quarry near the town of Picton. The second dike, exposed near the northwest corner of Varty Lake, was examined by Hon (1970), who concluded that this dike is a micaceous kimberlite. The Picton and Varty Lake dikes are the first post-Paleozoic ultramafic igneous rocks reported from the stable Paleozoic platform of southern Ontario. This paper provides basic information concerning the petrography, mineralogy, and mineral chemistry of these two occurrences and includes an estimate of the time of emplacement resulting from the application of the K–Ar and paleomagnetic techniques.

General geology

Both the Picton and Varty Lake dikes intrude limestones of the Ordovician Black River and Trenton groups. The Picton dike is exposed in the limestone quarry (44°03′N, 77°08′W) of the Lake Ontario Cement Company Limited located 4 km north of Picton. Host Trenton limestones are heavily jointed in the quarry but with no apparent vertical displacement across joints. The Picton dike occurs in one such joint exposed on three

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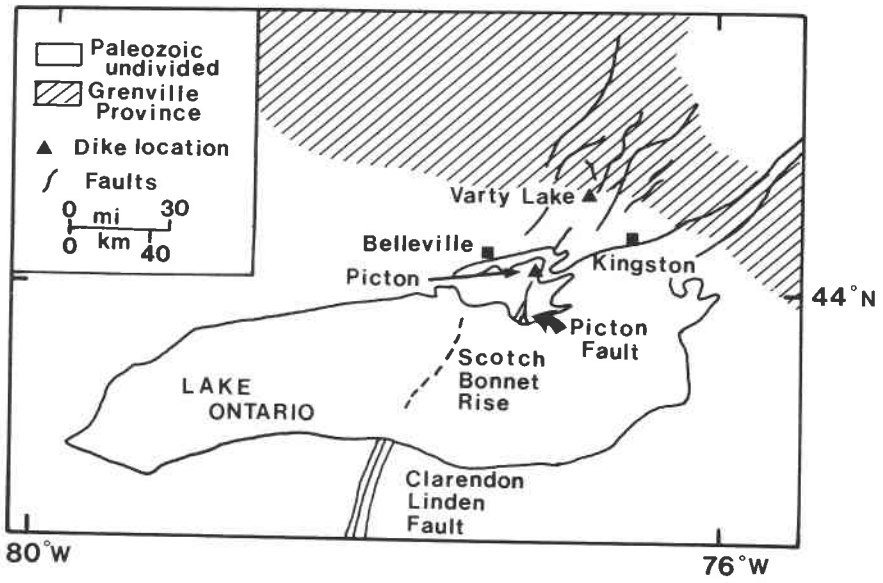


FIG. 1. Geological map of the Lake Ontario region showing the locations of the Picton dike and the Varty Lake dike. Faults within the Grenville Province and adjacent Paleozoic platform are taken from Baer *et al.* (1978) and Forsyth (1981). The Picton Fault, the Scotch Bonnet Rise, and the Clarendon–Linden Fault are taken from Liberty (1960), Hutchinson *et al.* (1979), and Fletcher and Sykes (1977), respectively.

bench levels and on the floor of the quarry. The dike is less than 0.2 m wide on the pit floor and on one of the benches and appears to pinch out to the southeast. To the northwest it attains its maximum width of 0.7 m on the upper bench level. The dike trends 105°, with dip ranging from vertical to 77°S.

The Varty Lake dike is intermittently exposed over a distance of 150 m in five prospect pits near the northwest corner of Varty Lake (44°25′N, 76°49′W). In the most complete exposure, the dike is 1.5 m wide and is vertical, with a strike of 120°. The limestone host rocks are flat lying in the area but beds have been tilted up about 15° on either side of the dike. Both dikes trend approximately normal to north-northeast-trending regional faults (Fig. 1).

Analytical procedures

Selected inclusion-free samples were crushed and analysed by the X-ray fluorescence method for whole-rock and trace-element compositions. The mineral chemistry was examined with a model 400 MAC electron microprobe and all data were processed using Magic IV matrix corrections.

The K–Ar dating procedure is identical to that described by Hayatsu and Carmichael (1977). Two large samples from the Picton dike and one from Varty Lake were crushed to 20 mesh and split in two. One portion was used untreated for a regular whole-rock method, and the other half was treated with 10% HCl to remove carbonate minerals before K and Ar determinations. A phlogopite mineral separate was also prepared from the Varty Lake dike for K–Ar determination.

Paleomagnetic measurements were made upon 12 oriented cores from the Picton dike and 8 oriented cores from the Varty Lake dike. In addition, a "baked contact test" (Everitt and Clegg 1962) was carried out on host limestones. Five cores were taken at varying distances from the dike–limestone contact at Picton, and eight cores at Varty Lake. Remanence measurements on the dike samples were performed using a Schonstedt spinner magnetometer and on the limestone host rocks using a cryogenic magnetometer. Stepwise alternating field (AF) and thermal demagnetizations were performed using Schonstedt GSD-5 and TSD-1 models.

Petrography

Picton dike

The Picton dike is distinctly heterogeneous, with a coarse-grained green to gray central region and a fine-grained pale green marginal zone up to 4 cm wide symmetrically disposed on either side, in contact with limestone. The dike contains fragments of limestone but the only ultramafic inclusion found was a glimmerite nodule. The glimmerite nodule is ovoid (1 cm × 0.5 cm) with a rounded outline similar to that of olivine phenocrysts that are present in the central part of the dike. Several phases of magma injection and brecciation are indicated by narrow sinuous stringers consisting of olivine and serpentine, typical of the central part of the dike, that penetrate into the marginal zone.

The central region of the Picton dike consists of 40–50 modal percent olivine and serpentinized phenocrysts and rare phlogopite phenocrysts in a fine-grained serpentine–calcite groundmass that includes phlogopite, perovskite, chlorite, acicular apatite, and blocky spinels (Table 1). Irregular remnants of fresh olivine (Fig. 2*a*) are present in some samples but most olivines are recognized by rounded serpentine pseudomorphs generally rimmed by minute grains of perovskite and magnesian–aluminous titanomagnetite.

Spinels occur with three texturally distinct habits in the Picton dike. Type 1 spinels are euhedral grains of magnesian–aluminous chromite less than 10 μm in size. These occur as inclusions in olivine or its serpentinized equivalent and are included within phlogopite grains of the glimmerite nodule. Type 2 spinels occur as crystal aggregates with atoll textures (Fig. 2*b*) that are present throughout the groundmass and rarely at the margins of rounded olivine pseudomorphs. They also are present in the angular interstitial regions of the glimmerite

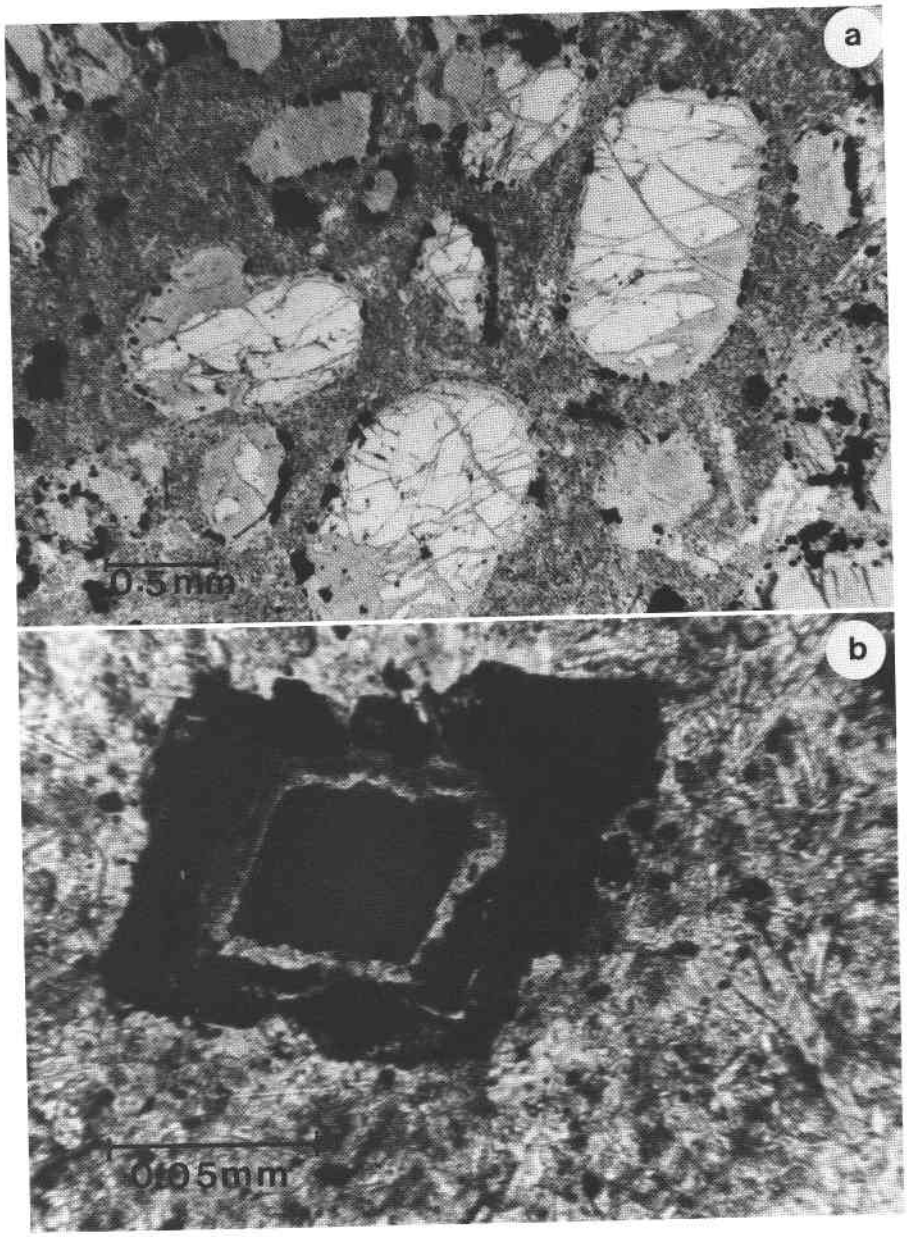


FIG. 2. Photomicrographs in plane-polarized light of a sample from the Picton dike. (*a*) Partly serpentinized olivine phenocrysts in a fine-grained groundmass of calcite, serpentine, spinel, and apatite. (*b*) Type 2 spinel with the atoll texture.

TABLE 1. Modal composition of the Picton and Varty Lake dikes

	Picton dike		Varty Lake dike	
	PI-8	DI	3–4	SD1-1
Olivine phenocrysts	4	—	—	—
Serpentinized phenocrysts	42	48	26	22
Phlogopite phenocrysts	<1	<1	20	24
Groundmass calcite	20	24	10	8
Opaque	9	11	4	5
Perovskite	<1	2	6	5
Serpentine and chlorite	19	11	28	32
Groundmass phlogopite	6	4	6	4

nodule. The type 2 spinels are up to 40 μm in diameter and consist of a relatively large central grain separated from a discontinuous outer zone of magnesian–aluminous titanomagnetite by a continuous reddish translucent intermediate region of an unidentified Fe–Mg–Al–Ti silicate. The larger central regions are composed of a small kernel of magnesian–aluminous chromite surrounded by titaniferous pleonaste. Type 3 spinels are euhedral grains of magnesian–aluminous titanomagnetite occurring with perovskite as a discontinuous rind surrounding rounded olivine and serpentinized olivine phenocrysts.

Calcite occurs throughout the groundmass of the Picton dike as aggregates of prismatic crystals associated with phlogopite and apatite in a serpentine matrix. Calcite and apatite needles commonly penetrate central regions of serpentine. These features are considered to have formed during late-stage crystallization of the dike. A second variety of calcite, occurring as randomly oriented laths having corroded margins, is more problematic. Lath-shaped calcite crystals have been observed replacing melilite in the Alnö ultramafic dike (von Eckermann

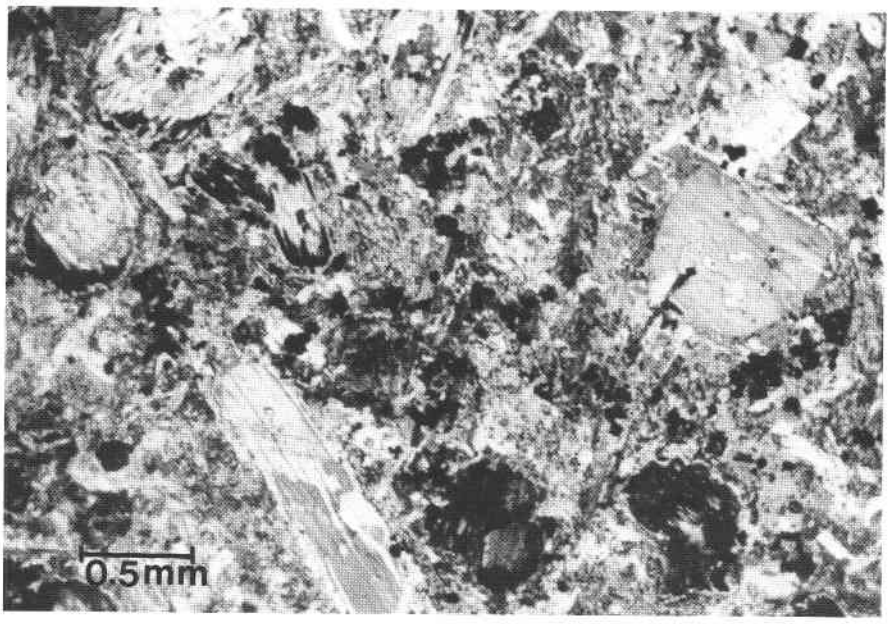


FIG. 3. Photomicrograph in plane-polarized light of a sample from the Varty Lake dike showing phenocrysts of phlogopite and pseudomorphs of olivine in a fine-grained groundmass of spinel, apatite, chlorite, calcite, and serpentine.

1948) and in several other ultramafic dikes (Watson 1967; Platt and Mitchell 1982). Relict melilite or possible alteration products of melilite such as garnet (Platt and Mitchell 1982) have not been observed in the Picton dike. The fine-grained marginal zones of the dike consist almost entirely of unoriented calcite laths with extremely sharp contact with adjacent unaltered wall rock. These relations further suggest that the lath-shaped calcite has an igneous origin. A third variety of calcite, consisting of fine-grained aggregates with angular margins, may represent recrystallized limestone fragments. The glimmerite nodule consists of interlocking phlogopite grains up to 1 mm in diameter with rare angular interstitial areas that contain type 2 spinels and ilmenite in a barite–calcite matrix. Apart from the obvious difference in the relative abundance of phlogopite the only other difference between the glimmerite nodule and the host dike concerns the titanium minerals. Perovskite occurs throughout the matrix of the dike, whereas ilmenite is present only in the barite–calcite matrix of the nodule. Identical chemistry of the spinels and phlogopite within both the dike and the nodule suggests that the glimmerite nodule is of cognate origin.

Varty Lake dike

The Varty Lake dike also has two distinct zones: a fine-grained marginal zone with rare phlogopite phenocrysts nearest the wall rocks on either side, and a much coarser central region with abundant phlogopite phenocrysts. The abundance of phlogopite distinguishes the Varty Lake dike from the Picton dike (Table 1). Samples from the coarse-grained central region contain between 20 and 30% phlogopite phenocrysts up to 3.0 mm in size and a lesser amount of serpentinized phenocrysts. The groundmass consists of phlogopite, spinel, chlorite, apatite needles, and massive to euhedral perovskite aggregates all set in a serpentine–calcite matrix (Fig. 3). The phlogopite phenocrysts optically exhibit two and often three concentric zones. Some phenocrysts have kink bands, possibly indicating some degree of deformation during emplacement. The bipyramidally terminated outline of some serpentinized phenocrysts suggests the former presence of olivine. However, the rounded edges of most of these serpentinized phenocrysts imply mechanical abrasion or absorption of original olivine during emplacement.

All three spinel types found in the Picton dike occur in the Varty Lake dike. Type 1 spinels, magnesian–aluminous chromite, are included in phlogopite phenocrysts as well as in serpentinized olivine phenocrysts. Type 2 spinels are smaller and are more poorly developed than in the Picton dike. Further, the central region of the atolls is magnesian–aluminous chromite that lacks the zone of titaniferous pleonaste, and a serpentine zone separates the inner zones from the outermost magnesian–aluminous titanomagnetite zone. Type 3 spinels occur at the margins of serpentinized olivine phenocrysts and as very fine grains throughout the groundmass. Pleonaste was not observed in any of the Varty Lake samples. Perovskite occurs as red translucent crystals throughout the groundmass and in irregular clusters at the outer margin of type 2 spinels. The narrow fine-grained marginal zone of the Varty Lake dike consists of discrete chromite grains, type 1 spinels, and perovskite in a serpentine matrix. Phlogopite and olivine phenocrysts are rare. Hon (1970) described globular cognate nodules consisting of Fe–Ti oxides, carbonate, and ovoid serpentine pseudomorphs. These features were not seen in the present study and presumably are uncommon.

Whole-rock analyses

Samples for whole-rock analyses were taken from the central part of both dikes. Major- and trace-element analyses are given in Table 2 together with published data from related rocks for comparison. Both dikes are characterized by low SiO_2 and Al_2O_3 and high MgO. Mg/(Mg + Fe) (Fe as total iron) and K/(K + Na) ratios are high. The Picton dike has higher CO_2 than the Varty Lake dike, reflecting higher calcite content. Although most of the calcite is believed to be of igneous origin, the high CaO and CO_2 contents may be partly the result of limestone contamination.

Most trace elements determined are within the range obtained from kimberlites (Dawson 1980; Wedepohl and Muramatsu 1979). Ba, Sr, and Co contents of both dikes are

TABLE 2. Chemical analyses of the Picton and Varty Lake dikes and comparison with kimberlites and kimberlitic rocks

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
SiO_2	26.0	29.8	22.7	33.4	35.2	—	—
TiO_2	1.52	3.34	5.86	2.52	2.32	—	—
Al_2O_3	2.94	4.55	3.09	5.09	4.4	—	—
Fe_2O_3	5.34	9.35	8.47	6.57	—	—	—
FeO	4.27	5.22	7.54	4.18	9.8*	—	—
MnO	0.15	0.22	0.21	0.17	0.11	—	—
MgO	23.4	23.8	23.8	23.8	27.9	—	—
CaO	14.5	8.80	11.8	9.06	7.6	—	—
Na_2O	0.02	0.04	0.27	0.44	0.32	—	—
K_2O	0.34	1.14	0.94	3.19	0.98	—	—
H_2O^+	8.50	7.70	6.22	7.25	7.4	—	—
H_2O^-	1.10	0.70	0.76	0.36	—	—	—
P_2O_5	0.65	1.24	0.68	0.44	0.7	—	—
CO_2	11.20	3.20	7.24	3.41	3.3	—	—
Total	99.93	99.10	99.58	99.88	100.03	—	—
				Trace element (ppm)			
Nb	170	334	—	—	—	110	32–450
Zr	172	303	—	—	—	240	84–700
Y	14	40	—	—	—	13	4–75
Sr	2642	1323	—	—	—	632	40–1900
Rb	35	38	—	—	—	58	0–350
Pb	14	20	—	—	—	8.5	0.9–50
Zn	63	121	—	—	—	85	15–287
Cu	20	17	—	—	—	85	10–300
Ni	772	317	—	—	—	1160	710–1600
Co	229	158	—	—	—	87	35–130
Cr	1326	1270	—	—	—	1380	550–2900
Ba	2078	2055	—	—	—	808	137–1970
V	220	229	—	—	—	98	4–75
S	828	494	—	—	—	2180	200–8800

*Total iron as FeO.

(1) Picton dike.

(2) Varty Lake dike.

(3) Bachelor Lake dike, Bachelor Lake, Quebec (Watson 1967).

(4) Glenwood dike, Ithaca, New York (Martens 1924).

(5) Average major-element content of kimberlite (Dawson 1980).

(6) Average trace-element content of kimberlite (Wedepohl and Muramatsu 1979).

(7) Range of trace-element content of kimberlite (Dawson 1980).

higher than most kimberlites, and the Ni content of the Varty Lake dike is lower.

Mineral chemistry

Phlogopite

The three textural varieties of phlogopite in the Picton dike (phenocryst, matrix, and glimmerite nodule) have consistent chemical differences in their TiO_2 and BaO contents (Fig. 4, Table 3). The phlogopite phenocrysts are uniform in composition and are characterized by high TiO_2 (3.04–3.47 wt.%), high Cr_2O_3 (0.85–1.15 wt.%), and low BaO (0.28–0.40 wt.%). Central parts of the phlogopites in the glimmerite nodule have compositions similar to that of the phenocryst phlogopites except for a lower Cr_2O_3 content (<0.42 wt.%). Phlogopites in the glimmerite nodule exhibit a chemical zonation from the centre to the margin that is enhanced with decreasing grain size. Smaller grains in the nodule do have optically distinct marginal zones that have higher BaO contents (analysis 3, Table 3). Small phlogopite platelets in the groundmass of the Picton samples are distinguished from phenocryst and glimmerite phlogopites by their low TiO_2 contents (0.22–0.88 wt.%) and by extremely high BaO (up to 6.17 wt.%, Fig. 4).

Phlogopite phenocrysts of the Varty Lake dike are composed of four chemically distinct concentric zones (Fig. 5). The central region, zone 1, is titanium phlogopite without detectable BaO. Zone 2, the next outward, is enriched in Cr_2O_3 (0.64–1.19 wt.%) with minor BaO content. Zone 3 is a band near the outer margin of the grain having extreme BaO enrichment. BaO content varies from 6.78 to 1.50 wt.% at the outer edge of zone 3. The last optically distinct region, zone 4, is a narrow poikilitic overgrowth that is chemically similar to groundmass phlogopite from the central part of the dike, characterized by low BaO and TiO_2 contents. Groundmass phlogopite from the fine-grained marginal zone of the dike has higher BaO contents than found in groundmass phlogopite from the centre of the dike.

The presence of high-barian phlogopite is a distinguishing feature of these two dikes. Phlogopites from both dikes also show similar chemical trends. BaO varies inversely with K_2O, MgO, and SiO_2 and varies directly with Al_2O_3 contents. There is no significant correlation between BaO and TiO_2 contents (Fig. 4). The antipathetic relation between K + Na and Ba (Fig. 6*a*) and the positive correlation between Ba and $Al^{IV}-Al^{VI}$ (Fig. 6*b*) indicate the replacement of K (and Na) by Ba in the interlayer site and the replacement of Si by Al in tetrahedral sites.

Spinel

Magnesian–aluminous chromites in types 1 and 2 of the Picton dike have nearly identical chemistries (Table 4). They are characterized by high Al_2O_3 (10–12.5 wt.%) and moderate TiO_2 contents (2.2–3.0 wt.%) and by high $Mg/(Mg + Fe^{2+})$ ratios (0.57–0.61). Type 1 spinels of the Picton dike are further characterized by narrow compositional zoning involving a decrease in Cr_2O_3 and an increase in Al_2O_3 from the centre to the marginal parts of grains (Table 4). X-ray intensity distribution scans of type 2 spinels reveal that titanium pleonaste and magnesian–aluminous chromite have a chemically sharp and distinct interface. The titanian pleonaste of type 2 spinels of the Picton dike is identical in composition to those in the glimmerite nodule. The magnesian–aluminous titanomagnetites of type 2 and type 3 spinels are also similar in chemistry. They have high MgO (13.8–15.3 wt.%) and Al_2O_3 (15.1–16.3 wt.%) and high $Mg/(Mg + Fe^{2+})$ ratios (0.54–0.55).

The magnesian–aluminous chromite of type 1 and type 2 spinels and the magnesian–aluminous titanomagnetite of type 2 and type 3 spinels of the Varty Lake dike are similar in chemistry to those in the Picton dike (Table 4).

The compositions of spinel phases of both dikes are plotted in the reduced spinel prism (Fig. 7 *a* and *b*). Spinel compositions are well within the range of spinels in kimberlites (Haggerty 1975; Mitchell 1979). The high $Mg/(Mg + Fe^{2+})$ ratios (0.45–0.55) of the magnesian–aluminous titanomagnetites are characteristic features of the late-stage spinels from kimberlites and are distinguished from those of lamprophyres and carbonatites (Mitchell 1979).

Olivine, ilmenite, and perovskite

Olivine is only slightly zoned, with the Mg/(Mg + Fe) ratio decreasing from 0.90 to 0.89 from cores to margins (Table 5). These values are within the range for olivines from kimberlites and associated rocks (Boyd and Clement 1977). Ilmenite occurs only in the glimmerite nodule from the Picton dike and possesses low MgO and low Cr_2O_3 (Table 6). These values are lower than those typically found in kimberlites (Dawson 1980).

TABLE 3. Representative electron microprobe analyses of phlogopite

	(1)	(2C)*	(3M)*	(4)	(5^{I})†	(6^{II})†	(7^{III})†	(8^{IV})†	(9)	(10)
SiO_2	38.88	37.65	35.96	32.63	39.02	38.30	31.75	38.04	38.84	37.81
TiO_2	3.47	2.78	2.91	0.72	2.30	2.20	1.84	1.23	0.64	0.24
Al_2O_3	14.10	15.54	17.07	19.04	13.24	14.00	19.18	13.39	14.45	16.78
Cr_2O_3	0.85	0.00	0.00	0.09	0.12	0.95	0.00	0.00	0.00	0.00
FeO‡	4.79	6.22	5.70	6.31	6.52	5.86	4.76	4.88	4.43	5.35
MnO	0.19	0.00	0.00	0.31	0.00	0.00	0.30	0.00	0.00	0.08
MgO	21.99	20.90	20.60	21.59	21.81	22.11	22.06	24.46	24.55	24.37
CaO	0.00	0.00	0.07	0.29	0.03	0.03	0.00	0.03	0.09	0.00
BaO	0.28	0.00	1.13	6.33	0.00	0.00	6.78	0.21	0.00	3.14
Na_2O	0.12	0.10	0.15	0.10	0.69	0.47	0.13	0.36	0.23	0.00
K_2O	10.20	10.32	9.50	8.27	9.27	9.60	7.68	9.82	9.94	9.59
Total	94.87	93.51	93.09	95.68	93.00	93.52	94.48	92.42	93.17	97.36
				Atomic proportions on basis of 22 oxygens						
Si	5.605	5.520	5.321	4.917	5.741	5.609	4.818	5.621	5.654	5.406
Ti	0.376	0.306	0.324	0.082	0.254	0.242	0.210	0.137	0.070	0.026
Al	2.396	2.685	2.977	3.380	2.296	2.416	3.430	2.332	2.479	2.827
Cr	0.097	0.000	0.000	0.011	0.014	0.110	0.000	0.000	0.000	0.000
Fe‡	0.578	0.763	0.705	0.795	0.802	0.718	0.604	0.603	0.539	0.640
Mn	0.023	0.000	0.000	0.040	0.000	0.000	0.039	0.000	0.000	0.010
Mg	4.725	4.567	4.543	4.849	4.783	4.827	4.989	5.387	5.327	5.194
Ca	0.000	0.000	0.001	0.047	0.005	0.005	0.000	0.005	0.014	0.000
Ba	0.016	0.000	0.065	0.374	0.000	0.000	0.403	0.012	0.000	0.176
Na	0.034	0.057	0.043	0.029	0.197	0.133	0.038	0.103	0.065	0.000
K	1.876	1.930	1.793	1.589	1.740	1.793	1.486	1.851	1.846	1.749
Total	15.726	15.828	15.782	16.113	15.832	15.853	16.017	16.051	15.994	16.028

*C and M are, respectively, core and margin of single grains.
†I, II, III, and IV, are, respectively, zone 1, zone 2, zone 3, and zone 4 of phenocryst phlogopite (see text).
‡FeO as total iron.
(1) Phenocryst phlogopite of the Picton dike.
(2, 3) Phlogopite of a glimmerite nodule of the Picton dike.
(4) Groundmass phlogopite of the Picton dike.
(5–8) Phenocryst phlogopite of the Varty Lake dike.
(9) Groundmass phlogopite of centre part of the Varty Lake dike.
(10) Groundmass phlogopite of fine-grain marginal zone of the Varty Lake dike.

Perovskite (Table 6) in the Picton dike and perovskite in the Varty Lake dike have FeO contents higher than and comparable to, respectively, those of perovskites contained in kimberlite (Dawson 1980). The unidentified reddish brown Fe–Mg–Al–Ti silicate mentioned in the petrographic section (above) is probably a hydrous silicate, based on the low totals in its microprobe analysis (Table 5). Preliminary X-ray analyses suggest that this phase has a serpentine-like structure. A detailed mineralogical study of this phase is under investigation and will be reported elsewhere.

K–Ar dating

The results of the potassium–argon analyses are shown in Table 7. For the Varty Lake dike, the whole-rock, acid-treated whole-rock, and phlogopite samples yielded concordant results. The phlogopite age is identical to that reported by Hon (1970) for an extract of phlogopite that was analysed by M. Shafiqullah when allowance is made for the new K–Ar constants (Steiger and Jäger 1977). It is concluded that the age of the Varty Lake dike is 170 Ma. However, it should be noted that the three ages given in Table 7 may not be independent: the potassium and argon contents of the whole-rock and acid-treated samples appear to be largely controlled by the phlogopite in these samples. As a consequence, the presence of excess argon in the phlogopite cannot completely be ruled out.

For the Picton dike, the ages of two acid-treated samples agree with each other, whereas the whole-rock samples yielded a greater discrepancy in age and both are older than the acid-treated samples. Since the calculated age of the dissolved component is not drastically different from the geological age of the Trenton Limestone, the whole-rock ages were probably affected by carbonates incorporated from the host-rock limestone. Therefore, the average of the two HCl-treated ages of 173 Ma is considered the age of the dike.

Paleomagnetic results

Varty Lake dike

The natural remanent magnetization (NRM) directions from the Varty Lake dike samples were dispersed; no specimen from the eight independently oriented samples of the dike reached stable end points with either AF or thermal demagnetization (Webster 1982). The limestone samples at this site, on the other hand, exhibited greater stability. The NRM directions were either south and shallow initially or were inclined steeply downward initially and became south and shallow with AF demagnetization. Collectively, the directional results from the four limestone samples examined were similar to what would be expected from North America over the broad time interval from Middle Ordovician to Late Permian (Table 1 of Van der Voo 1982). No obvious difference in characteristic

TABLE 4. Representative electron microprobe analysis of spinel

	(1)	(2)	(3)	(4)	(5)	(6)
TiO_2	3.02	2.38	1.50	13.29	3.22	12.73
Al_2O_3	14.25	10.59	50.02	15.69	11.10	4.05
Cr_2O_3	40.19	48.32	0.00	0.02	46.14	1.66
Fe_2O_3*	12.98	9.81	18.94	33.19	9.04	44.60
FeO*	15.77	16.50	9.40	22.19	19.89	25.32
MnO	0.34	0.19	0.18	0.42	0.20	0.67
MgO	13.58	12.51	20.70	15.32	10.88	11.57
Total	100.13	100.30	100.74	100.12	100.47	100.60
	Atomic proportions on basis of 4 oxygens					
Ti	0.072	0.058	0.030	0.315	0.080	0.326
Al	0.534	0.405	1.562	0.583	0.427	0.162
Cr	1.011	1.239	0.000	0.001	1.192	0.045
Fe^{3+}	0.311	0.240	0.378	0.787	0.222	1.142
Fe^{2+}	0.419	0.447	0.208	0.585	0.544	0.720
Mn	0.009	0.005	0.004	0.011	0.005	0.019
Mg	0.644	0.605	0.818	0.719	0.530	0.587
$Mg/(Mg + Fe^{2+})$	0.61	0.58	0.80	0.55	0.49	0.45

*Fe_2O_3 and FeO are calculated on magnetite–ulvöspinel basis (Carmichael 1967).
(1) Mg–Al chromite in the Picton dike.
(2) Mg–Al chromite in the Picton dike.
(3) Mg–Fe pleonaste in the Picton dike.
(4) Mg–Al titanomagnetite in the Picton dike.
(5) Mg–Al chromite in the Varty Lake dike.
(6) Mg–Al titanomagnetite in the Varty Lake dike.

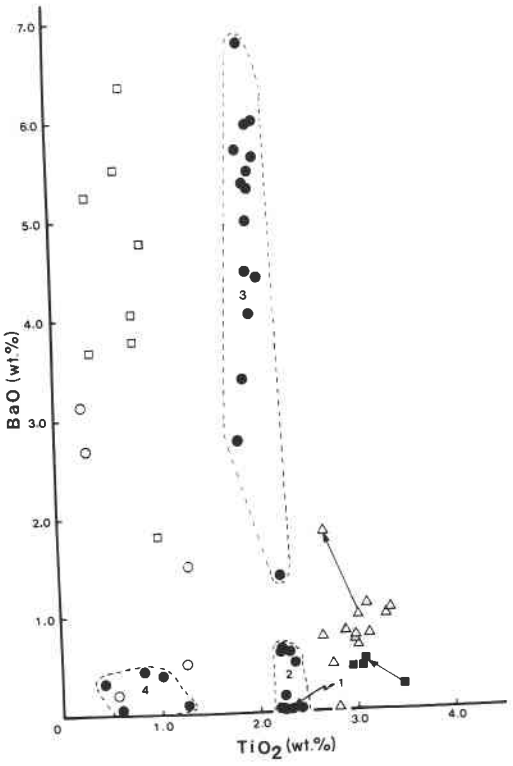


FIG. 4. BaO vs. TiO_2 in phlogopites from the Picton and Varty Lake dikes. Picton phlogopites are represented by solid squares (phenocrysts), open squares (groundmass), and open triangles (glimmerite nodules). Varty Lake dike phlogopites are represented by solid circles (phenocrysts) and open circles (groundmass). Dotted areas 1, 2, 3, and 4 correspond to compositional zones in Varty Lake phlogopite phenocrysts. Arrows indicate core–margin relationships of single grains.

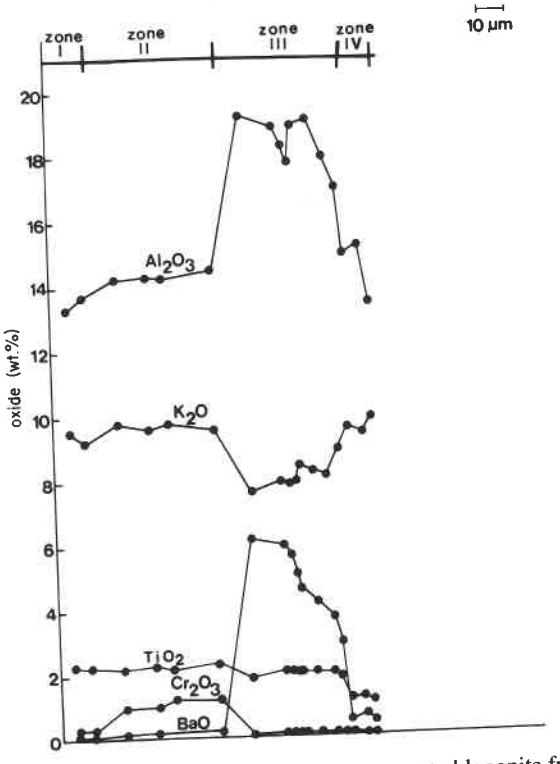


FIG. 5. Chemical zoning profile of phenocryst phlogopite from the Varty Lake dike.

remanence directions can be perceived in samples from within and outside the probable baked zone of the dike; this may indicate a very low emplacement temperature for the dike, since the dike failed to impress a Jurassic magnetization on the limestone adjacent to the contact. Because the NRM in the dike

TABLE 5. Representative electron microprobe analyses of olivine and unidentified Fe–Mg–Al–Ti silicate

	(1C)*	(2M)*	(3)
SiO_2	41.19	41.11	23.87
TiO_2	0.00	0.00	19.03
Al_2O_3	0.00	0.00	12.99
Cr_2O_3	0.04	0.06	0.24
FeO†	9.70	11.22	24.63
MnO	0.05	0.11	0.04
MgO	47.94	46.98	5.69
CaO	0.09	0.18	0.61
Na_2O	0.00	0.00	0.11
K_2O	0.00	0.00	0.13
Total	99.01	99.66	87.34
Atomic proportions on basis of 4 oxygens			
Si	1.016	1.016	0.737
Ti	0.000	0.000	0.472
Al	0.000	0.000	0.442
Cr	0.001	0.001	0.006
Fe†	0.200	0.231	0.636
Mn	0.001	0.002	0.001
Mg	1.760	1.727	0.262
Ca	0.002	0.005	0.020
Na	0.000	0.000	0.006
K	0.000	0.000	0.005

*C = core; M = margin.
†Total iron as FeO.
(1, 2) Olivine phenocryst of the Picton dike.
(3) Average of 18 analyses.

TABLE 6. Representative electron microprobe analyses of perovskite and ilmenite

	(1)	(2)	(3)
TiO_2	54.98	53.09	49.05
Al_2O_3	0.60	0.31	0.00
Cr_2O_3	0.24	0.19	0.14
FeO*	5.38	2.11	47.35
MnO	0.02	0.12	0.50
MgO	0.54	0.07	0.39
CaO	32.04	37.15	0.39
Na_2O	0.34	0.52	0.00
K_2O	0.02	0.02	0.00
La_2O_3	—	1.10	—
Ce_2O_3	—	2.90	—
Pr_2O_3	—	0.09	—
Nd_2O_3	—	1.13	—
Er_2O_3	—	0.05	—
Nb_2O_5	—	0.73	—
Total	94.16	99.58	97.87
Fe_2O_3†			2.47
FeO†			45.13

*Total iron as FeO.
†Fe_2O_3 and FeO calculated by Carmichael's (1967) method.
(1, 2) Peroviskites of the Picton and Varty Lake dikes, respectively.
(3) Ilmenite of the glimmerite nodule of the Picton dike.

rocks is magnetically unstable the paleomagnetic method fails to confirm the emplacement age of the Varty Lake dike.

Picton dike

The Picton dike samples, in contrast, possess a very stable NRM. The initial NRM of all samples is of reversed polarity. All samples exhibited similar patterns on AF demagnetization; with the data plotted in the form of orthogonal demagnetization diagrams, the samples show linear decay to the origin (Fig. 8) after initial removal of soft components in fields of the order of 10 mT. The specimens examined by thermal demagnetization exhibited similar directional behaviour, with linear segments generated over the temperature range 300–400 to 550 or 575°C. All specimens were completely unblocked after the 600°C run (Fig. 8).

The remanence in the samples of Trenton Limestone, collected at varying distances from the dike contact, was analysed exclusively by the AF demagnetization technique. The magnetic hardness of the limestone remanence varies directly with proximity to the dike contact. Median destructive fields of the NRM in terms of distance from the dike–limestone contact, expressed in dike widths, are as follows: 100 mT at 0.1; 55 mT at 0.6; 10 mT at 1.25; 7.5 mT at 3.25; and 6 mT at 40. The characteristic remanence directions in the dike and limestone were obtained by principal component analysis following the method of Kirschvink (1980). These directional results are graphically displayed in Fig. 9. Statistical data derived from these results are given in Table 8.

Because the dike has imposed a thermally induced paleomagnetic signature on the contact rocks immediately adjacent to it (Fig. 9), it is certain that this remanence dates from the time of intrusion and subsequent cooling. Furthermore, the

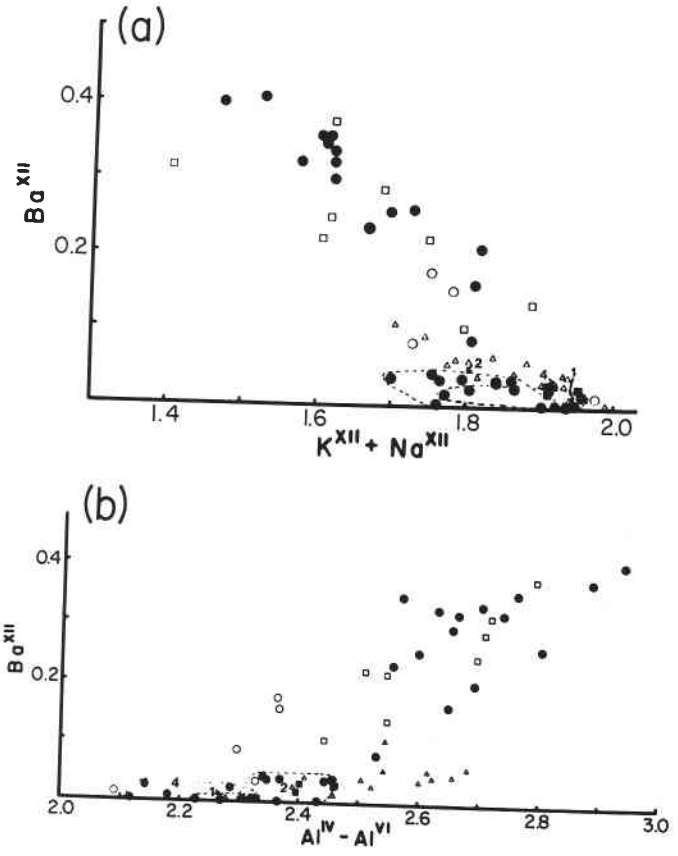


FIG. 6. (*a*) Ba^{XII} vs. $K^{XII} + Na^{XII}$ in phlogopites from Picton and Varty Lake dikes. Symbol conventions are as in Fig. 4. (*b*) Ba^{XII} vs. $Al^{IV} - Al^{VI}$ in phlogopites from Picton and Varty Lake dikes. Symbol conventions are as in Fig. 4.

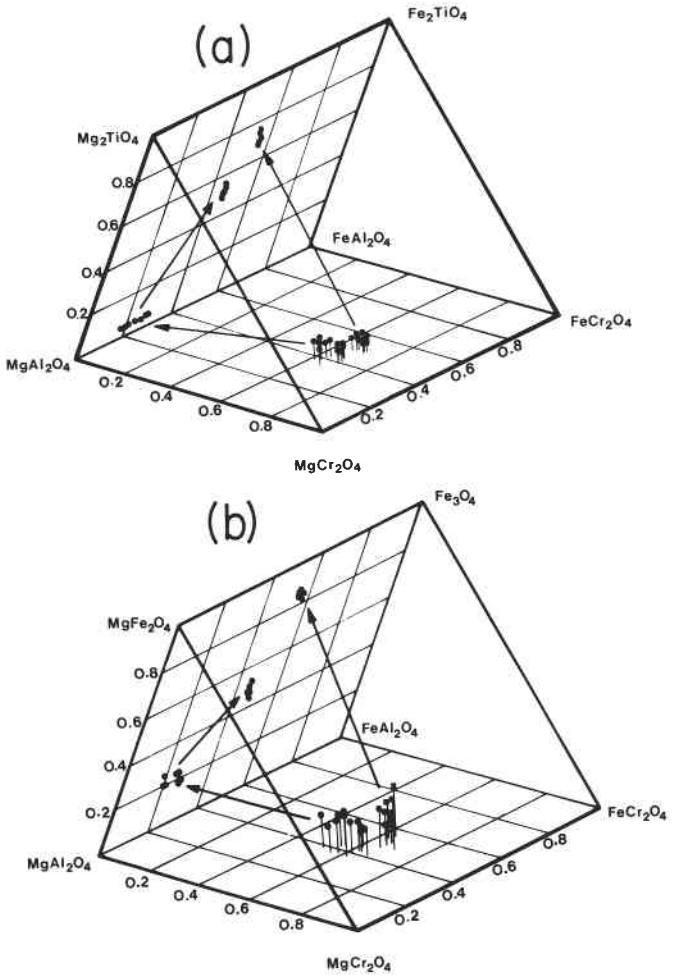


FIG. 7. Spinel compositions from the Picton and Varty Lake dikes plotted in the reduced spinel prisms. Solid circles indicate Picton spinels, whereas solid squares indicate Varty Lake spinels. Arrows indicate core−margin relationships in type 2 spinels.

characteristic remanence is coherent in direction with temperature in the 550−575°C range, implying an emplacement temperature in excess of 575°C. The unblocking temperatures obtained in laboratory time scales are likely to be comparable to the *in situ* blocking temperatures because of the rapidity with which the dike cooled, given its thickness at the point of sampling (cf. Fig. 17 of Dunlop 1981).

Although the paleomagnetic data enable a minimum estimate of the dike's emplacement temperature, its presumed rapid cooling history has prevented a reasonable estimate of its time of emplacement. The paleomagnetic directional result (Table 8) represents a spot reading of the geomagnetic field in some past reversed polarity field. The virtual geomagnetic pole derived from the directional result of Table 8 is plotted in Fig. 10 with the stable North American, Late Carboniferous to present apparent polar wander path (APWP) also shown for reference. It can be seen that the Picton dike pole does not fall on this path or fall close to the few mid-Paleozoic results from the interior of North America (Irving 1979; Van der Voo 1982). Although local tectonic effects cannot be ruled out as an explanation for the discrepancy, nonaveraging of paleosecular variation, as implied by the probable rapid cooling manifested by the low within-site scatter of directions (Table 8), is the most probable cause for the lack of coincidence with any segment of the

TABLE 7. K−Ar analyses of the Picton and Varty Lake dike samples

	K content (wt.%)	Radiogenic Ar (cm^3/g)	Age (Ma)
Varty Lake dike			
Whole-rock	1.306	9.074×10^{-6}	170
HCl-treated	1.443	9.907×10^{-6}	169
Phlogopite	7.48	52.20×10^{-6}	171
Picton dike			
Sample A			
Whole-rock	0.452	3.540×10^{-6}	191
HCl-treated	0.662	4.683×10^{-6}	173
Sample B			
Whole-rock	0.476	3.855×10^{-6}	197
HCl-treated	0.666	4.762×10^{-6}	175
Carbonates*	0.1044	2.081×10^{-6}	451
Trenton Limestone	0.0404	1.661×10^{-6}	

*K and Ar contents were calculated from the following relation:

$$X^{\text{whole rock}} = fX^{\text{dissolved}} + (1-f)X^{\text{residue}}$$

where X is K or Ar content and f is the fraction dissolved.

APWP. The individual results upon which the master APWP is based, on the other hand, represent results that have integrated local paleomagnetic fields over periods ranging from 10^3 to 10^6 years. Based on a minimum departure argument from the master APWP, the most probable time of emplacement is Cenozoic or Jurassic. The K−Ar results (see above) have, however, resolved this ambiguity and both dikes have ages near the Early−Middle Jurassic boundary in the time scale given by Van Hinte (1978).

Petrological and tectonic discussion

Based on textural relations, both dikes exhibit similar crystallization sequences. The early crystallization of magnesian−aluminous chromite followed by olivine and Ba-poor phlogopite depleted the residual magma in K_2O, Al_2O_3, and MgO and enriched it in BaO. Crystallization of high-barian phlogopite of both dikes may reflect BaO enrichment and increasing Ba/K in the residual magma. An alternative explanation for the BaO enrichment of phlogopite phenocrysts of the Varty Lake dike invokes the mixing of high-BaO-bearing magma or fluid with low-BaO-bearing magma in which BaO-poor phlogopites have already crystallized prior to the mixing event. A mixing process, however, seems unlikely in the Varty Lake dike because a systematic zoning trend was observed in all the phlogopite phenocrysts studied and no indication of a mixing process is evident from chemistry of type 1 spinels, which presumably crystallized simultaneously with phlogopite phenocrysts.

Both dikes have fine-grained marginal zones that exhibit sharp contacts with the central parts of the dikes, indicating that these marginal zones represent injections of volatile-rich fractionated liquid (Clement 1975) rather than flow differentiation products (Hon 1970). Primary magmas of the dikes probably had high H_2O-CO_2 contents similar to those of kimberlite. The rarity of xenocrysts and cognate nodules implies a limited degree of crystal fractionation of the magmas. This is supported by the primitive chemistry of both dikes, involving high concentrations of the compatible elements (Ni, Co, and Cr), and by the high $Mg/(Mg + Fe^T)$ ratios.

Spinels with atoll texture and chemistry similar to the type 2 spinel in this study have been reported from the kimberlites of Peuyuk, Somerset Island, Canada (Mitchell and Clarke 1976),

TABLE 8. Summary of remanence directional data from the Picton dike

N	D (°)	I (°)	R	k	α_{95} (°)	Long. (°W)	Lat. (°N)	dp (°)	dm (°)
12	198	−68	11.975	434	2.1	22	76	$2\frac{1}{2}$	$3\frac{1}{2}$
14*	200	−66	13.946	242	2.6	14	76	$3\frac{1}{2}$	4

NOTES: N is the number of specimens; D and I are the declination and inclination of the mean direction and R is the resultant; k and α_{95} are Fisher's precision and circle of confidence ($P = 0.05$); dp and dm are the semi-angles of the oval of confidence about the pole position given in terms of longitude and latitude.

*This entry contains two directions from Trenton Limestone within the baked zone of the dike.

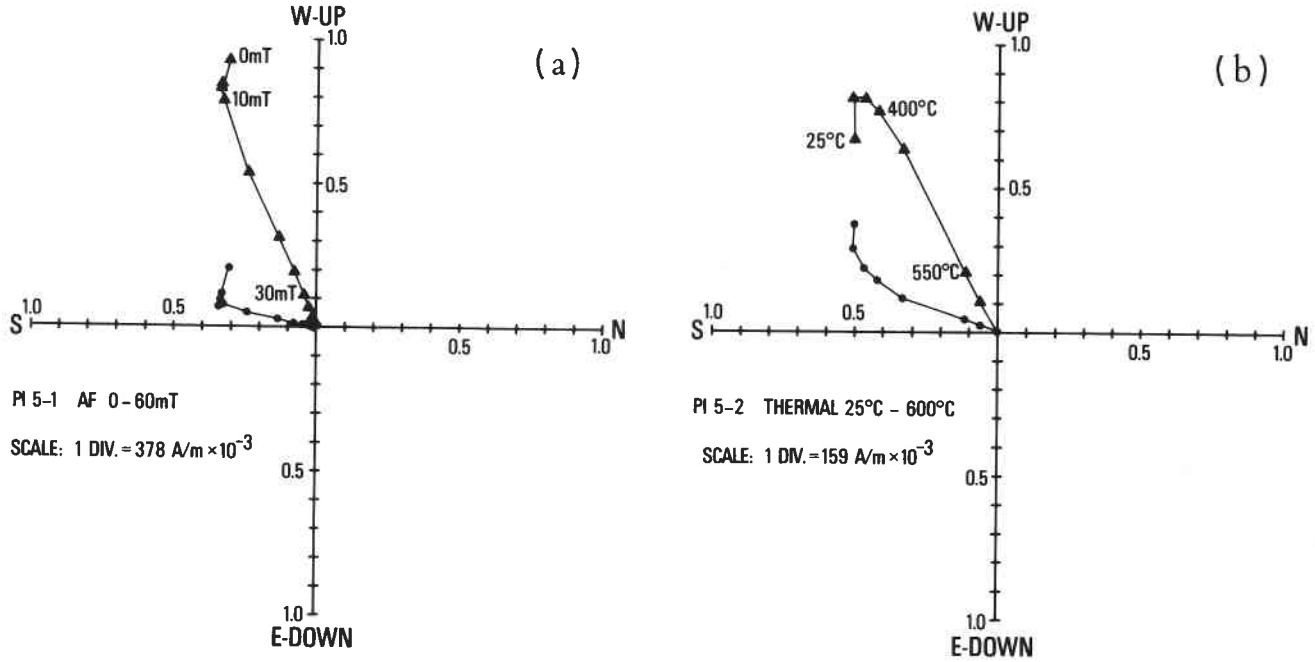


FIG. 8. Vector diagrams for two specimens from Picton dike sample 5. Dots represent points on the horizontal plane and triangles correspond to projections on the north−south vertical plane. In both AF (a) and thermal (b) treatments, the diagrams show straight-line segments decaying to the origin.

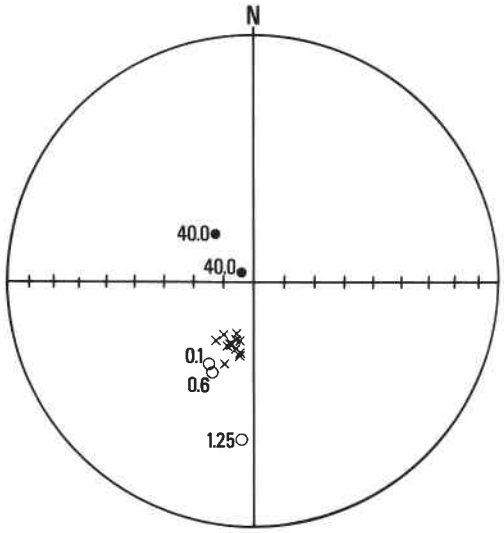


FIG. 9. Equal-area projection showing characteristic remanence directions (×'s) obtained from the Picton dike samples. Open and closed circles are characteristic remanence directions obtained from Trenton Limestone samples collected at varying distances from the dike. The numbers beside the open and closed circles indicate the distance (expressed in dike widths) of the sample from the dike. The ×'s and open circles represent directions on the upper hemisphere; closed circles represent directions on the lower hemisphere.

and the De Beers mine, South Africa (Pasteris 1983). These authors have suggested a resorption origin for the atoll texture. Central grains of the atoll texture in the Picton and Varty Lake dikes commonly show euhedral outlines. The unidentified Fe−Mg−Al−Ti silicate associated with atoll spinels in the Picton dike is probably a product of the reaction related to resorption and serpentinization of titanomagnetite.

Despite numerous similarities of petrography and mineralogy, some significant differences remain between the Picton and Varty Lake dikes. Firstly, phlogopite phenocrysts are dominant in the Varty Lake dike yet are rare in the Picton dike (Table 2). Secondly, type 2 spinels in the Picton dike have a margin of titaniferous pleonaste around the central chromite-rich core. Pleonaste is not present in the Varty Lake dike. These differences might have resulted from different trends of chemical evolution of the magmas controlled by crystallization of phlogopite (Boctor and Boyd 1983; Pasteris 1983). Crystallization of relatively large amounts of phenocryst phlogopites in the Varty Lake dike would have resulted in depletion of Al_2O_3 in the residual liquid. In contrast, less extended crystallization of phenocryst phlogopite would have resulted in a residual magma sufficiently aluminous for pleonaste crystallization in the Picton dike.

Based on the mineralogical classification of kimberlite as proposed by Skinner and Clement (1979), the Picton dike is a calcite−serpentine kimberlite and the Varty Lake dike a

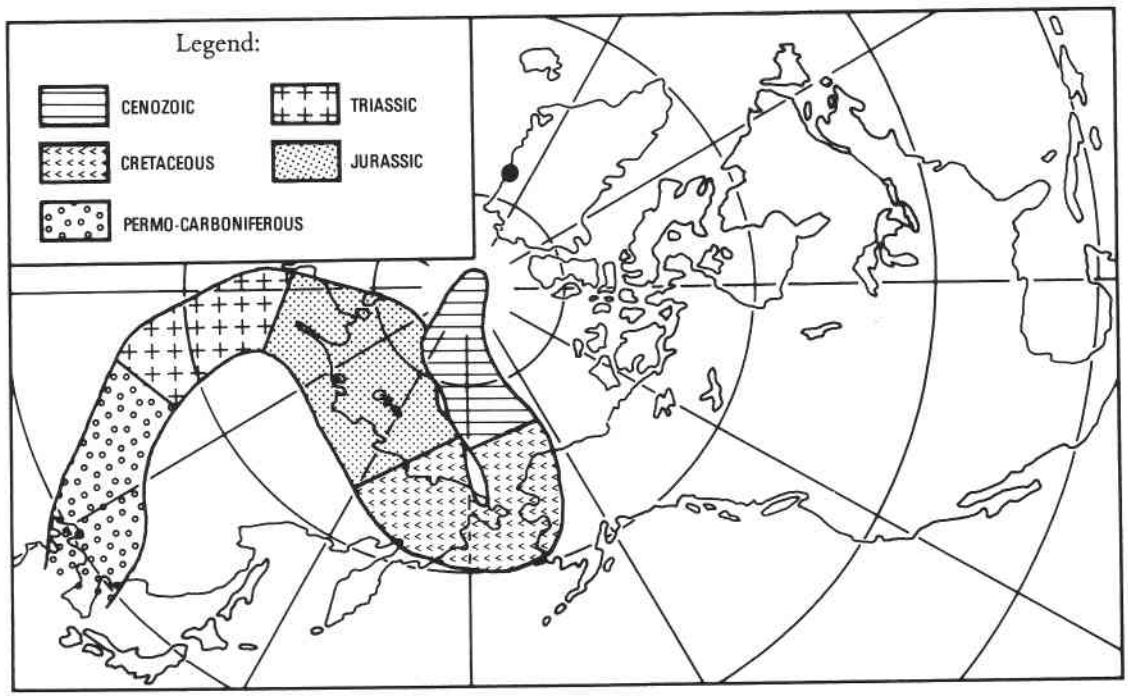


FIG. 10. Polar envelope for stable North America, Late Carboniferous time to the present, generalized from Irving (1979). The south pole obtained from the Picton dike is represented by the solid dot. Note that the Picton pole does not fall on the polar path; it is closest to the average pole during either Cenozoic or Jurassic time.

phlogopite–calcite–serpentine kimberlite. However, the high BaO contents of the phlogopites in both dikes are not characteristic of kimberlites in which phlogopites have BaO contents less than 0.7 wt.% (Gaspar and Wyllie 1982). On the contrary, high-barian trioctahedral mica is more typical of leucite-bearing rocks (Thompson 1977; Holm 1982; Birch 1980), nephelinite (Velde 1979; Mansker *et al.* 1979), carbonatite (Gaspar and Wyllie 1982), and monticellite peridotite (Wendlandt 1977). Most of these rocks have feldspathoids (leucite, melilite, or nepheline), which were not found in the Picton and Varty Lake dikes. In the absence of feldspar and (or) feldspathoids these dikes cannot be classed as lamprophyres (Streckeisen 1978). However, because of the absence of picroilmenite, Cr diopside, and pyrope, the Picton and Varty Lake dikes cannot be classed as kimberlite in the classification of Mitchell (1979, 1983). In summary, petrographical and chemical features of the Picton and Varty Lake dikes preclude simple petrological classification, as both dikes have aspects indicative of kimberlitic and lamprophyric affinities. The magnesian–aluminous titanomagnetites, which are characteristic of kimberlite (Mitchell 1979), and the high-barian phlogopite seem to be unique features of these dikes.

In a comprehensive compilation of seismic, geomagnetic, geological, and structural features related to the Western Quebec Seismic Zone, Forsyth (1981) presented convincing evidence that the Central Metasedimentary Belt of the Grenville Province (Wynne-Edwards 1972) extends in a southwesterly direction and forms the basement to southeastern Ontario, Lake Ontario, and western New York State. The Picton dike occurs within 2 km of the projected trace of a major fault indicated by Kay (1937) and mapped by Liberty (1960). The Varty Lake dike is close to the projected trace of one of several north-northeast-trending faults in the Grenville basement. The faults at Picton are subparallel to, but offset laterally from, the Scotch Bonnet Rise (Hutchinson *et al.* 1979), which appears to be a continuation of the Clarendon–Lindon Fault (Fletcher and Sykes 1977). Geological and geophysical evidence (Hutchinson *et al.* 1979) suggests that the Scotch Bonnet Rise and the Clarendon–Linden Fault are basement structures. Reactivation of these structures took place in the Ordovician and again in post-Devonian time (Fletcher and Sykes 1977). The association of kimberlites and alkaline rocks with basement faults, reactivated during the early stages of continental fragmentation, is common on a global scale (Sykes 1978). The spatial association of both the Picton dike and the Varty Lake dike with faults or the projected traces of faults (Fig. 1) suggests that the southern Ontario portion of this major basement fault system was reactivated in the Jurassic and served as a conduit for the products of deep-seated magmatism. The magmatism itself is probably a distal expression of continental rifting prior to the opening of the North Atlantic.

Acknowledgments

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